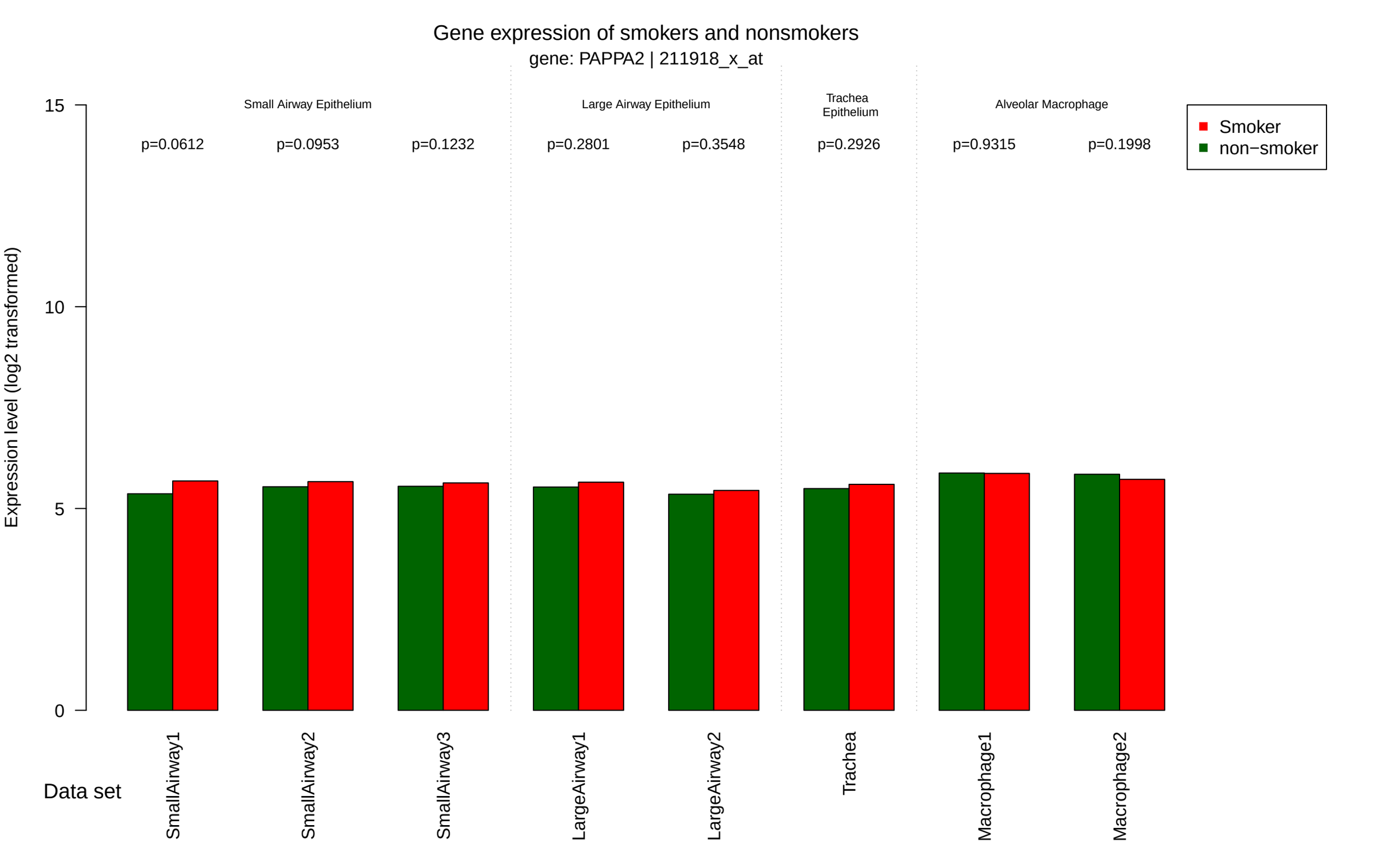
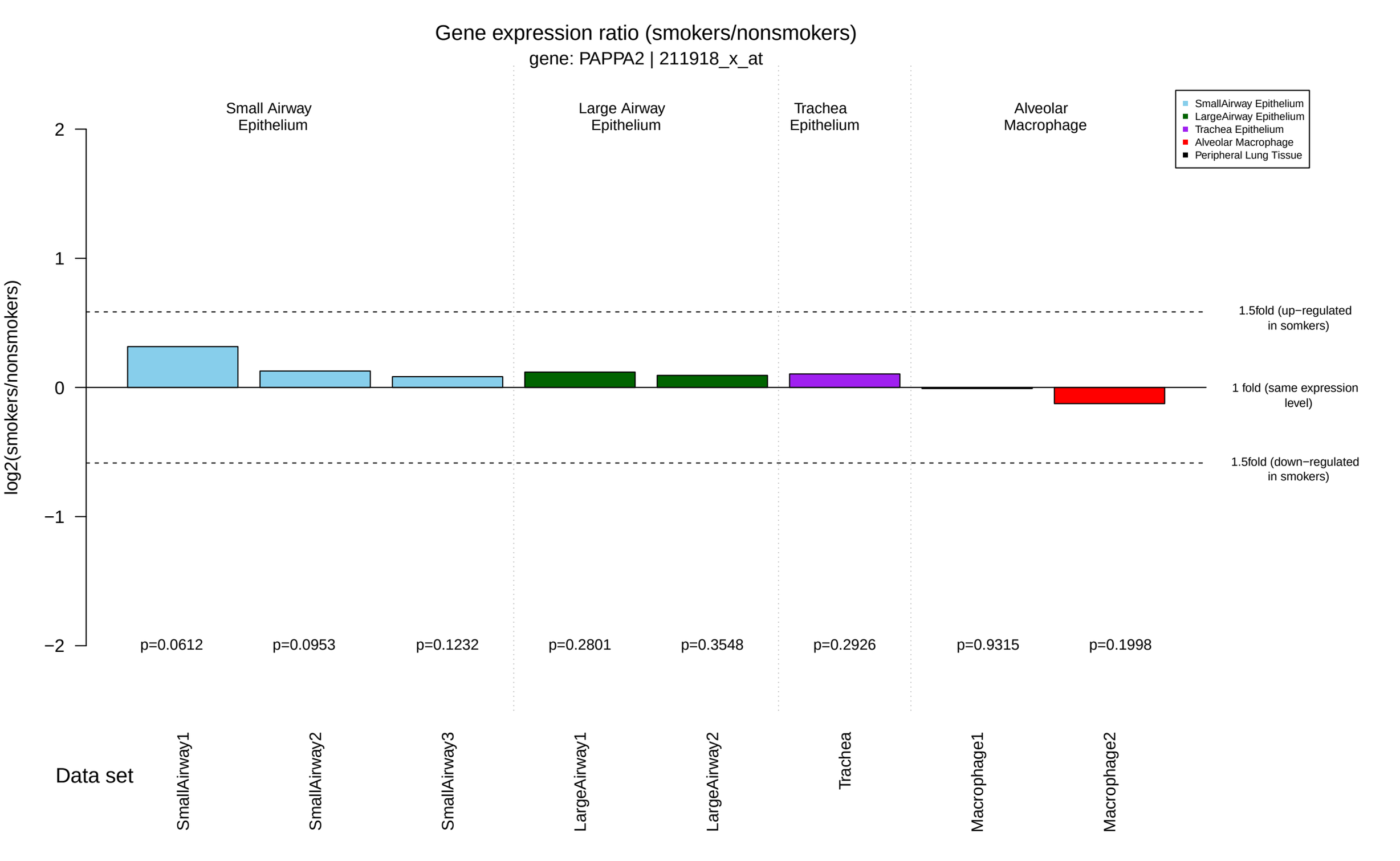
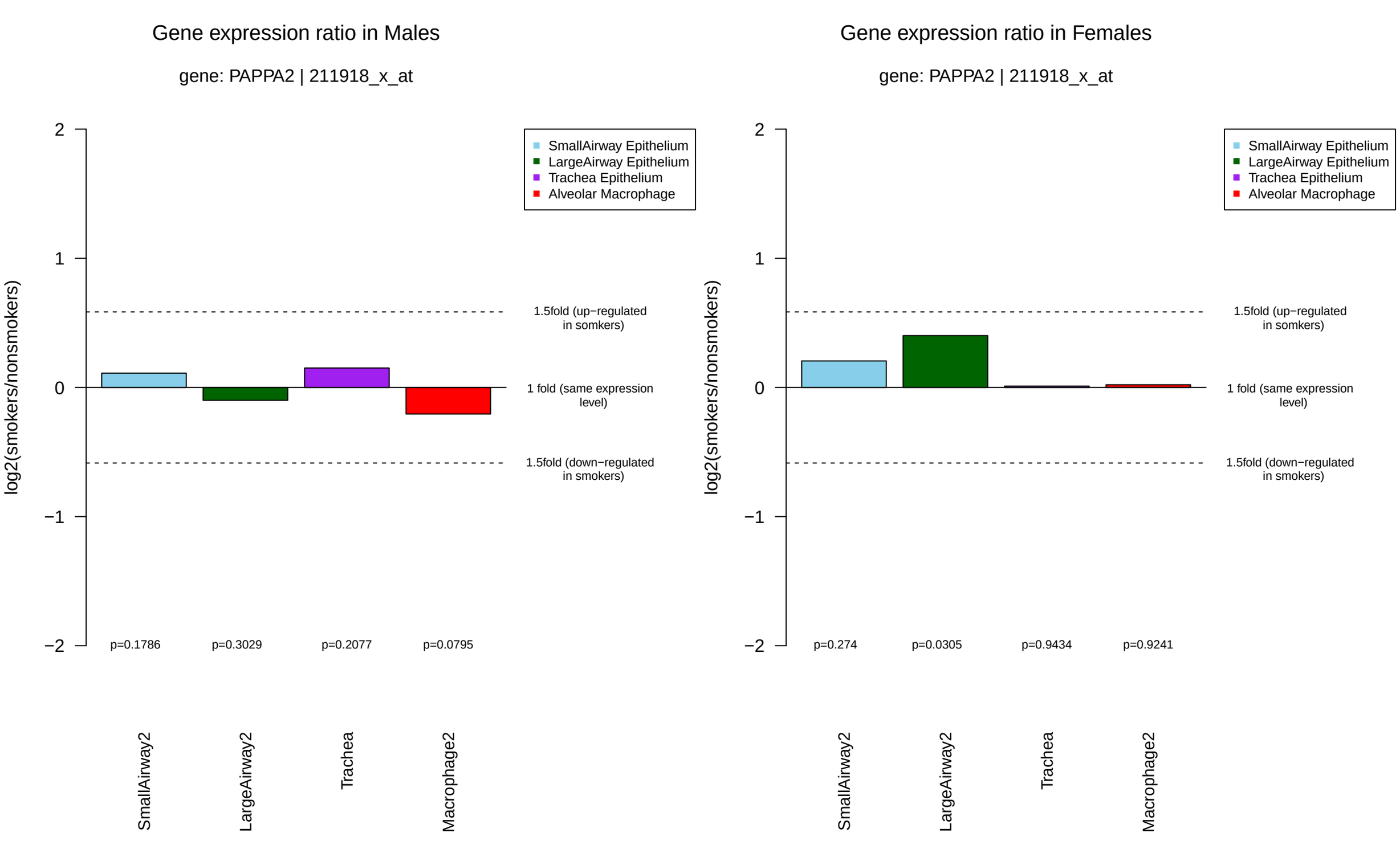
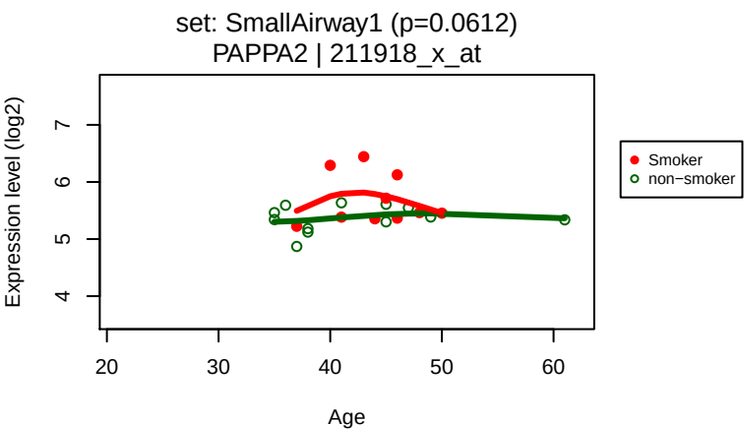
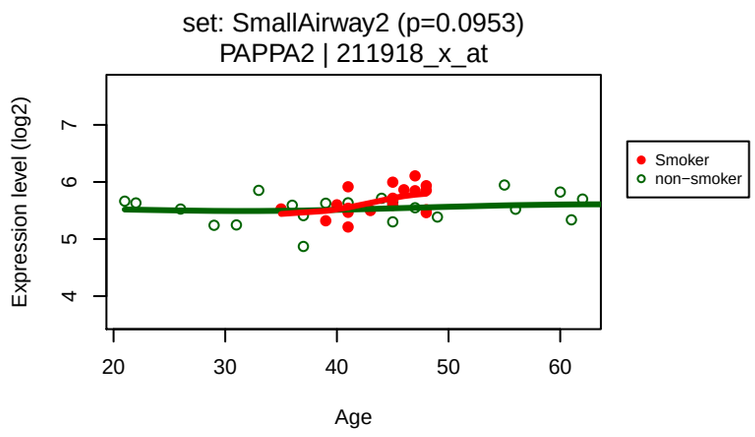
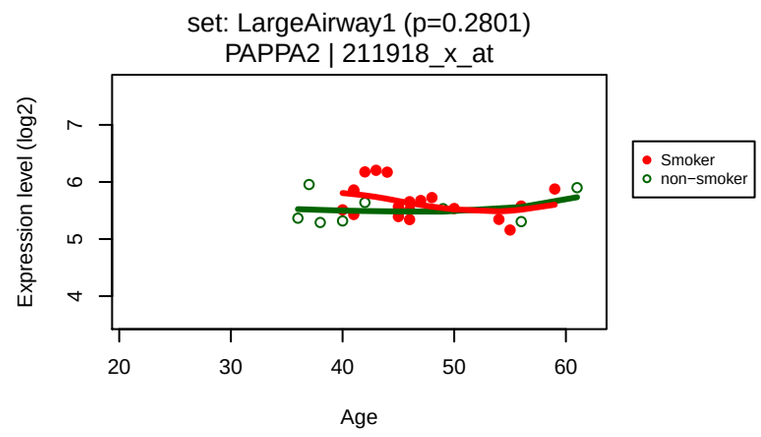

Gene expression of smokers and nonsmokers
gene: PAPPA2 | 211918_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0612
p=0.0953
p=0.1232
p=0.2801
p=0.3548
p=0.2926
p=0.9315
p=0.1998
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

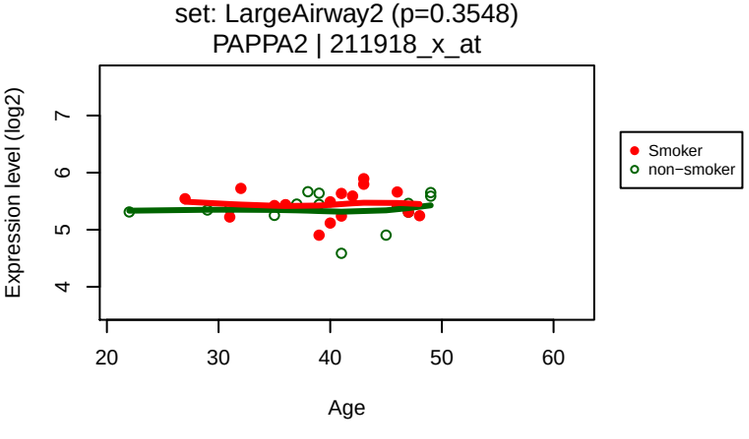

Gene expression ratio (smokers/nonsmokers)
gene: PAPPA2 | 211918_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0612
p=0.0953
p=0.1232
p=0.2801
p=0.3548
p=0.2926
p=0.9315
p=0.1998
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

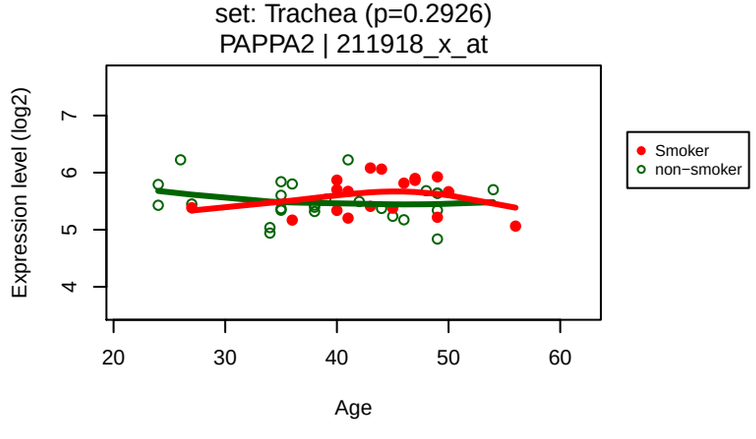

Gene expression ratio in Males
gene: PAPPA2 | 211918_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1786
p=0.3029
p=0.2077
p=0.0795
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PAPPA2 | 211918_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.274
p=0.0305
p=0.9434
p=0.9241
SmallAirway2
LargeAirway2
Trachea
Macrophage2

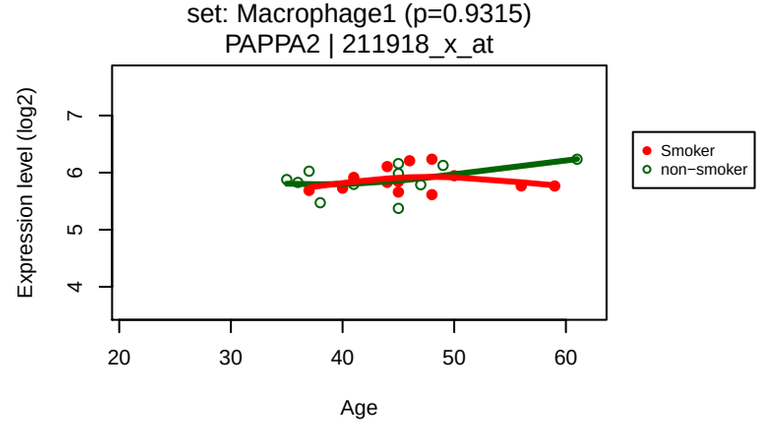

set: SmallAirway1 (p=0.0612)
PAPPA2 | 211918_x_at
Expression level (log2)
Age
Smoker
non−smoker

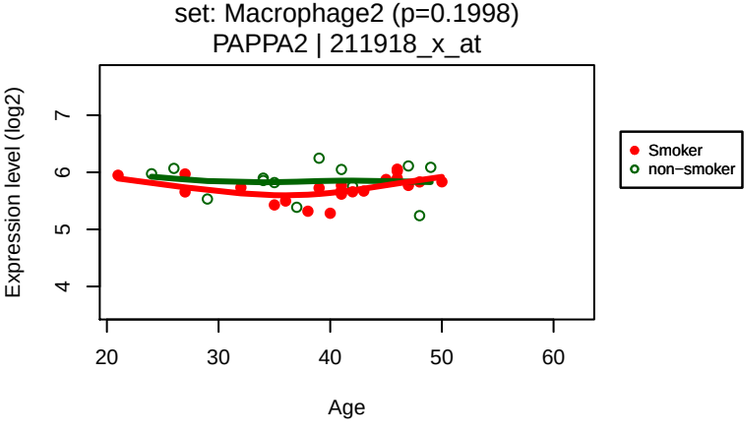

set: SmallAirway2 (p=0.0953)
PAPPA2 | 211918_x_at
Expression level (log2)
Age
Smoker
non−smoker

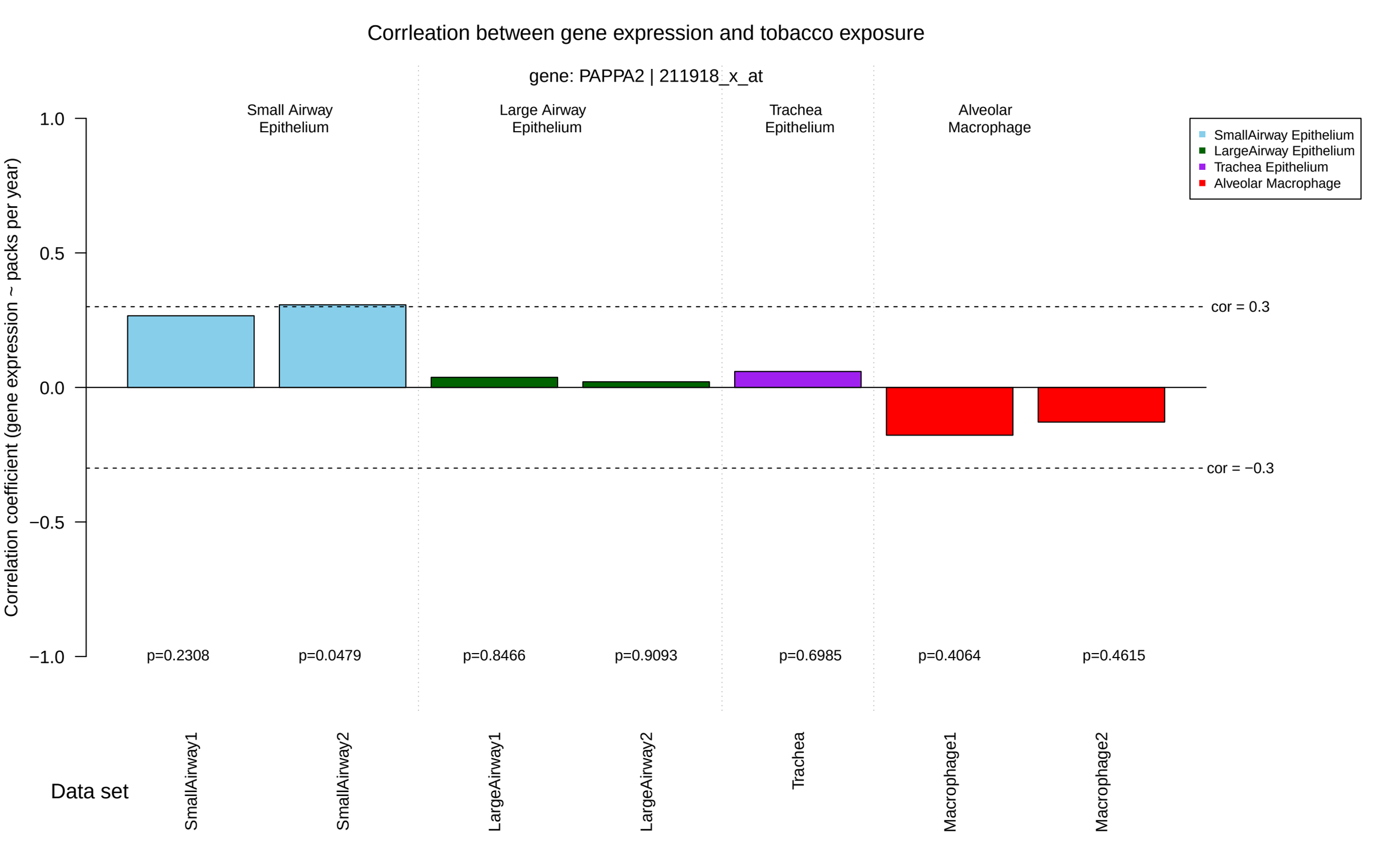

set: LargeAirway1 (p=0.2801)
PAPPA2 | 211918_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3548)
PAPPA2 | 211918_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.2926)
PAPPA2 | 211918_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.9315)
PAPPA2 | 211918_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1998)
PAPPA2 | 211918_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PAPPA2 | 211918_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2308
p=0.0479
p=0.8466
p=0.9093
p=0.6985
p=0.4064
p=0.4615
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2